Elliptic Billiards as Exact Poncelet Rotations: Monotonicity, Porisms, and the Clean-Family Obstruction

Route-A source-local certificate HCS-C275

1 September 2026

Abstract

For the billiard inside a confocal ellipse, restricted to positively oriented orbits with an elliptic caustic, we give the exact Jacobi covering and rigid rotation number. We prove strict monotonicity in both eccentricities, both rotation endpoints, and that reduced rotation p/q makes every orbit on the caustic curve a least-period-q Poncelet polygon. The qth restricted return is the identity with unit tangent derivative on a full periodic circle, obstructing the requested ordinary isolated-orbit product. A 613-cell receipt audits every convention. The theorem excludes hyperbolic caustics and makes no ambient unipotent or target spectral claim.

1 The frozen confocal owner

For $0 < \varepsilon < 1$ let

$$E(\varepsilon) = \left\{(x, y) : \varepsilon^2 x^2 + \frac{\varepsilon^2}{1-\varepsilon^2} y^2 = 1\right\}.$$

Its foci are $(\pm 1, 0)$. Fix $0 < f < e < 1$: $E(f)$ is the outer billiard boundary and $E(e)$ is an inner elliptic caustic. We keep the positive orientation and one-reflection clock. The corresponding Poncelet circle map is denoted $B_e^f : E(f) \to E(f)$. This is precisely the elliptic-caustic sector analyzed in the primary rotation-number sources [1, 2, 3]; hyperbolic caustics are not included.

Our convention is

$$F(\phi, e) = \int_0^\phi \frac{d\tau}{\sqrt{1 - e^2 \sin^2 \tau}}, \qquad K(e) = F(\tfrac{\pi}{2}, e),$$

so e is the Jacobi modulus (software parameter $m = e^2$). Define

$$\omega(e, f) = \arcsin \sqrt{\frac{e^2 - f^2}{e^2(1 - f^2)}}, \tag{1}$$

$$\rho(e, f) = \frac{F(\omega(e, f), e)}{2K(e)}. \tag{2}$$

Theorem 1 (Jacobi covering and rigid rotation)**.** *The period-one covering*

$$\pi_e^f(\theta) = \frac{1}{f}\left(-\operatorname{sn}(4K(e)\theta, e), \sqrt{1 - f^2}\, \operatorname{cn}(4K(e)\theta, e)\right) \tag{3}$$

satisfies

$$B_e^f \circ \pi_e^f = \pi_e^f \circ R_{\rho(e,f)}, \qquad R_\rho(\theta) = \theta + \rho.$$

In particular, (2) *is the rotation number.*

Proof. The identity $\operatorname{sn}^2 + \operatorname{cn}^2 = 1$ puts (3) on $E(f)$, and the $4K(e)$ period gives the deck map $\theta \mapsto \theta + 1$. Set $v = 2F(\omega, e)$. The Jacobi addition formulas, together with (1), show that the line through $\pi_e^f(\theta)$ and $\pi_e^f(\theta + v/(4K(e)))$ obeys

$$C^2 = A^2/e^2 + (1 - e^2)B^2/e^2$$

when written as $Ax + By + C = 0$. This is the dual equation for tangency to $E(e)$. Positive orientation selects the next intersection, proving the conjugacy; $v/(4K(e))$ is exactly (2). This also verifies the hypotheses and conclusion of the confocal covering theorem in [1]. ☐

2 Strict parameter rigidity and endpoints

The inverse form of (2) is especially effective:

$$\rho(e,f)=\ell \quad\Longleftrightarrow\quad f=R(\ell,e):=e\ \mathrm{cd}(2K(e)\ell,e), \qquad 0<\ell<\tfrac{1}{2}. \tag{4}$$

Theorem 2 (strict signs and the two endpoint values)**.** *Throughout* $0<f<e<1$*,*

$$\partial_e\rho(e,f)>0, \qquad \partial_f\rho(e,f)<0.$$

Moreover,

$$\lim_{f\to e^-}\rho=\lim_{e\to f^+}\rho=0, \qquad \lim_{f\to 0^+}\rho=\lim_{e\to 1^-}\rho=\tfrac{1}{2},$$

where the parameter not tending to its boundary is held fixed.

Proof. Let $u=2K(e)\ell$ and $\mathcal{E}(u,e)=\int_0^u \mathrm{dn}^2(t,e)\,dt$. Differentiating (4) gives

$$R_\ell=-\frac{2e(1-e^2)K(e)\,\mathrm{sn}(u,e)}{\mathrm{dn}^2(u,e)}<0$$

and

$$R_e=\mathrm{cd}(u,e)+\frac{\mathrm{sn}(u,e)}{\mathrm{dn}^2(u,e)}[\mathcal{E}(u,e)-2\ell\mathcal{E}(K(e),e)].$$

Since $\mathrm{dn}^2(t,e)$ strictly decreases on $(0,K(e))$, its average on $[0,u]$ exceeds its average on $[0,K(e)]$; the bracket and hence R_e are positive. Implicit differentiation of $f=R(\rho,e)$ yields $\rho_f=1/R_\ell<0$ and $\rho_e=-R_e/R_\ell>0$.

The two coalescing limits follow from $\omega\to 0$; $f\to 0$ gives $\omega\to\pi/2$ and $F\to K$. For fixed f and $e\to 1^-$, put $e'=\sqrt{1-e^2}$. Then $\cos\omega=O(e')$ and

$$0\le K(e)-F(\omega,e)=\int_\omega^{\pi/2}\frac{dt}{\sqrt{1-e^2\sin^2 t}}=O(1),$$

whereas $K(e)\sim\log(4/e')\to\infty$. Thus $F/K\to 1$ and the last limit is $1/2$. ☐

Theorem 3 (complete elliptic-caustic porism)**.** *If* $\rho(e,f)=p/q$ *with* $0<p/q<1/2$ *and* $\gcd(p,q)=1$*, every orbit on this caustic curve has minimal period* q*.*

Proof. In the covering coordinate, the qth iterate sends θ to $\theta+p$, a deck-equivalent point. If $0<k<q$ returned, then kp/q would be integral; coprimality would force $q\mid k$, a contradiction. The arbitrary starting value of θ produces the continuous Poncelet family, recovering the porism statement of [3]. ☐

3 The clean family and the Route-A boundary

The conjugacy gives more than a period count:

$$(R_{p/q})^q(\theta)=\theta+p, \qquad D_\theta(R_{p/q})^q=1.$$

Consequently $(B_e^f)^q$ restricted to this invariant circle is the identity, and its tangent derivative is one at every point. The fixed set is a circle, not a discrete collection. Hence the ordinary isolated-primitive-orbit product requested at A2, and a nondegenerate isolated-fixed-point denominator, are unavailable on the rational caustic. This does not rule out a separately defined Morse–Bott or Berry–Tabor regularization. It also does not determine the transverse multiplier or assert an ambient Jordan/unipotent form.

The receipt contains 32 rotation-formula cells, 192 covering and tangent-chord cells, 117 strict-monotonicity values, 96 endpoint values, 24 inverse-porism cases, 128 polygon vertices, and 24 restricted-return derivative cells. A producer-independent checker, 208 exact symbolic checks, fresh byte replay, and 24 repaired-hash mutations audit the package. Finite cells test conventions; the preceding arguments prove the parameter theorem.

Let Ω_f be the smooth bounded interior of $E(f)$. Its standard ambient Dirichlet operator is self-adjoint with compact resolvent on $L^2(\Omega_f)$, with

$$-\Delta_D, \qquad \mathcal{D}(-\Delta_D)=H^2(\Omega_f)\cap H_0^1(\Omega_f).$$

The unitary group $U(t) = \exp[-it(-\Delta_D)]$ uses continuous physical flight time. Complex conjugation C is antiunitary and obeys $CU(t)C = U(-t)$. This is a coherent ambient quantum billiard, but the frozen classical owner is the one-reflection Poincaré map. No same-clock quantum return has been constructed, and no theorem shows that such a return retains the fixed-caustic orbit phases and weights. Therefore A4 is only

$$\texttt{A4\_FORMAL\_HINT}.$$

No quantum spectrum is identified with a target divisor. The frozen scope is

$$\texttt{NO\_BAD\_EULER\_OR\_ROOT\_NUMBER}.$$

Its strict tuple is

$$\begin{gathered}\texttt{(A0\_FAIL,A1\_PASS\_ANALYTIC,A2\_FAIL,}\\ \texttt{A3\_FAIL,A4\_FORMAL\_HINT)}.\end{gathered}$$

The overall verdict is `ROUTE_A_REJECTED`; Route B is disabled. The theorem covers only the positive-orientation elliptic-caustic sector.

References

[1] H. E. Lomelí and J. D. Meiss, *Symmetry Reduction and Rotation Numbers for Poncelet maps*, arXiv:2309.08013v1 (2023), doi:10.48550/arXiv.2309.08013.

[2] R. Kołodziej, *The rotation number of some transformation related to billiards in an ellipse*, Studia Math. 81 (1985), 293–302, doi:10.4064/sm-81-3-293-302.

[3] S.-J. Chang and R. Friedberg, *Elliptical billiards and Poncelet's theorem*, J. Math. Phys. 29 (1988), 1537–1550, doi:10.1063/1.527900.